

Confidence Interval And Margin of Error.

Point Estimate

Definition - A single statistic (eg. sample mean $\bar{x}$, sample proportion $\hat{p}$) calculated from sample data to estimate an unknown population parameter (eg. population mean μ , population proportion p)

Example - If you survey 100 people and find the average age is 35, then 35 is the point estimate for the average age of the entire population

Limitation - It is a single value & does not convey how close it might be to the

true parameter, as it is subject to sampling error.

Confidence Interval

Defⁿ - An interval of values, derived from a point estimate and a margin of error, within which a true population parameter is expected to lie with a certain probability.

Formula = Point Estimate $\pm$ margin of error

Components

- Point Estimate - The center of interval (mean)
- margin of - How far the interval extends

0
Error from the point estimate
Depends on sample size,
variability & confidence level)

- Confidence Level - The probability
(eg. 90%, 95%) that the interval contains
the true parameter if repeated samples
were taken

OR

It tells us how sure we are that the
true value falls within the interval

- Example: A 95% CI of (32, 38) for
the average age means we are 95%
confident the true average age falls
between 32 & 38

Margin of Error -

The measure of how much the true value might be off from the estimate

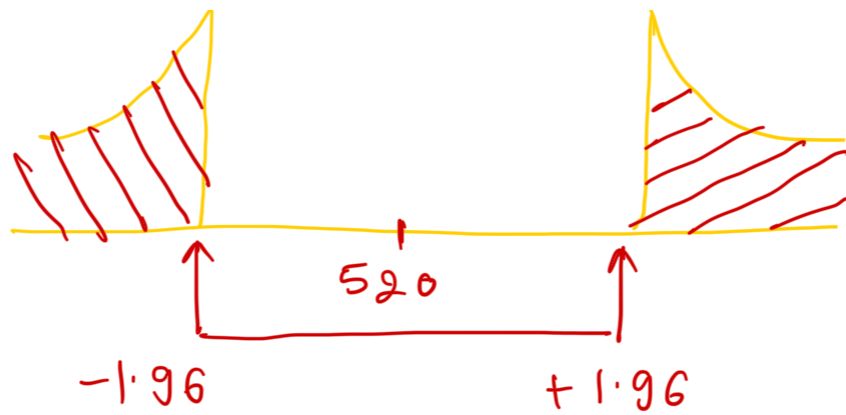
$$\text{Margin of Error} = z \times \frac{\sigma}{\sqrt{n}}$$

Example

On the verbal section of cat exam, the std deviation is known to be 100. A sample of 25 test takers has a mean of 520. Construct the 95% CI around the mean?

$$\bar{x} = 520, \sigma = 100, n = 25, CI = 0.95, \alpha = 0.05$$





Point Estimate $\pm$ Margin Error

$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$

$$\begin{aligned}\text{Lower CI} &= \bar{x} - z \frac{\sigma}{\sqrt{n}} = 520 - 1.96 \times \frac{100}{5} \\ &= 480.8\end{aligned}$$

$$\begin{aligned}\text{Higher CI} &= \bar{x} + z \frac{\sigma}{\sqrt{n}} = 520 + 1.96 \times \frac{100}{5} \\ &= 559.2\end{aligned}$$

Conclusion

I am 95% confident that the mean CAT score lies between 480.8 & 559.2